

E-COMMERCE BUSINESS PERFORMANCE ANALYSIS

1. CLIENT INFORMATION

- **Client** : Shopkart
- **Industry** : E-Commerce
- **Business Model** : B2C (Business-to-Consumer) Online Retail
- **Client Role** : Management Team of shopkart
- **Project Name** : E-Commerce Business Performance Analysis
- **Company Background** :

ShopKart is a fictional large-scale e-commerce marketplace that sells products across multiple categories through an online platform.

The company works with multiple sellers and serves customers across different cities and states. Customers can browse products, place orders, make payments, use offers or coupons, and receive deliveries through ShopKart's logistics network.

The business generates revenue through product sales and marketplace-related activities. Management monitors sales performance, customer activity, seller performance, product demand, order fulfillment, payments, returns, and delivery operations.
- **Current Business Situation** :

The business has grown significantly, but management is finding it increasingly difficult to understand where the business is performing well and where it is losing money or customers.

Management currently receives data from different business systems, but decision-making is becoming difficult because there is no single clear view of overall business performance.

The management team wants an analytical review of the available business data before making major operational and growth decisions.

2. BUSINESS PROBLEM

As the business owner, our main concern is that sales are growing, but we don't have enough visibility into what is actually driving the business performance. We want to understand what is happening across customers, products, sellers, orders, payments, deliveries, and returns.

Some of the questions and concerns raised by management are :

Sales

- Which periods are generating strong or weak sales?
- Are sales growth patterns consistent?
- Which categories and products contribute most to sales?
- Are discounts and offers helping sales or reducing revenue unnecessarily?

Customers

- Which customer groups generate the most business?
- Are new customers becoming repeat customers?
- Which locations have strong customer activity?
- Are customers becoming inactive after placing orders?
- How important are repeat customers to overall sales?

Products

- Which products are performing strongly?
- Which products have weak demand?
- Which categories generate the most revenue?
- Are some products frequently ordered but also frequently returned?

Sellers

- Which sellers contribute significantly to sales?
- Are some sellers generating unusually high cancellations or returns?
- Are seller performances consistent across different periods?

Orders

- How many orders are successfully completed?
- How significant are cancelled and returned orders?
- Are there particular periods or categories with unusual order behavior?

Payments

- Which payment methods are commonly used?
- Are certain payment methods associated with failed or cancelled orders?
- Are payment-related problems affecting completed sales?

Delivery

- Are orders being delivered within expected timelines?
- Are certain locations, sellers, or categories experiencing delivery problems?
- Could delivery performance be affecting customer experience?

Returns

- Which products or categories have high return activity?
- Are returns concentrated among particular sellers or customer segments?
- How much business activity is affected by returned orders?

3. BUSINESS REQUIREMENTS

Mandatory Requirements

Management expects the Data Analyst to:

1. Understand and document the available business data.
2. Examine overall sales and order performance.
3. Analyze customer purchasing behavior.
4. Analyze product and category performance.
5. Evaluate seller performance.
6. Examine order cancellations and returns.
7. Analyze payment-method usage and related order outcomes.
8. Evaluate delivery performance.
9. Identify important business trends and unusual patterns.

Questions Management Wants Answered

1. What is driving our overall sales performance?
2. Which products and categories are most valuable to the business?
3. Which customer segments contribute the most revenue?
4. Are customers returning to purchase again?
5. Which sellers are performing well and which require attention?
6. Where are cancellations and returns concentrated?
7. Are discounts and offers having a positive business effect?
8. Which locations perform strongly or poorly?
9. Are there delivery problems affecting orders?

Optional Requirements

If the available data supports it, management would also like:

- Customer segmentation
- Product performance ranking
- Seller comparison
- Regional performance comparison
- Monthly/quarterly trend analysis
- Discount and offer analysis
- Repeat-purchase analysis
- Return behavior analysis

4. BUSINESS OBJECTIVES

The objective of this project is to give management a clear, data-driven view of business performance.

Specifically, we want to:

- Improve visibility into sales performance.
- Understand customer purchasing behavior.
- Identify high-performing and underperforming products.
- Evaluate seller performance.
- Understand order cancellation and return patterns.
- Monitor delivery and operational performance.
- Identify areas where revenue may be lost.
- Support better business decision-making.
- Establish meaningful KPIs for regular management monitoring.
- Identify opportunities for sustainable business growth.

5. SCOPE

- The project includes analysis of:
- Sales performance
- Customer behavior
- Customer purchasing patterns
- Products
- Product categories
- Sellers
- Orders
- Order status
- Payments
- Discounts/offers
- Delivery performance
- Returns
- Geographic performance
- Business KPIs
- Trends and patterns
- Management-level insights

OUT OF SCOPE

The Data Analyst is not responsible for:

- Developing or modifying the e-commerce website.
- Developing mobile applications.
- Changing payment systems.
- Managing warehouse operations.
- Managing delivery employees.
- Negotiating with sellers.
- Creating marketing campaigns.
- Making changes directly inside production systems.
- Building machine-learning models unless separately requested.
- Financial auditing or statutory accounting.
- Making final business decisions on behalf of management.

The analyst's responsibility is to analyze the available information and provide evidence-based findings and recommendations.

6. DATA SOURCES

ShopKart will provide data from several business areas.

Customers

Contains information about customers registered on the platform.

Business purpose:

Used to understand customer demographics, locations, registration activity, and customer behavior.

Orders

Contains information about customer orders.

Business purpose:

Used to analyze order volume, order dates, order status, payment methods, and order-level sales activity.

Order Items

Contains individual products included within each order.

Business purpose:

Used to analyze product quantities, prices, discounts, and order-level product details.

Products

Contains information about products available on the platform.

Business purpose:

Used to analyze products, categories, pricing, and product performance.

Sellers

Contains information about sellers operating on the marketplace.

Business purpose:

Used to evaluate seller activity and performance.

Payments

Contains payment-related transaction information.

Business purpose:

Used to understand payment methods, payment status, and transaction behavior.

Deliveries

Contains order delivery information.

Business purpose:

Used to evaluate delivery timelines and delivery performance.

Returns

Contains information about returned orders or products.

Business purpose:

Used to understand return activity and identify return patterns.

Inventory

Contains product stock-related information.

Business purpose:

Used to understand product availability and inventory-related business activity.

7. EXPECTED DELIVERABLES

Management expects the following deliverables:

Data

- Original datasets preserved.
- Cleaned datasets.
- Documented data preparation process.

SQL Analysis

- Business-focused SQL queries.
- Analytical results addressing management questions.
- Relevant aggregations, joins, filtering, and advanced analysis where required.

Python Analysis

- Exploratory data analysis.
- Statistical summaries where appropriate.
- Visual analysis.
- Business-focused findings.

KPI Report

A defined set of important business KPIs covering areas such as:

- Sales
- Orders
- Customers
- Products
- Sellers
- Returns
- Delivery
- Payments

Dashboard

An interactive management dashboard showing important business metrics, trends, comparisons, and performance indicators.

Final Report

A business-oriented report containing:

- Major findings
- Important trends
- Business problems identified from the data
- Recommendations
- Potential business impact
- Limitations

Management Presentation

A concise presentation that allows management to understand the major findings without going through the complete technical analysis.

8. SUCCESS CRITERIA

The project will be considered successful if:

- Management can clearly understand overall business performance.
- Major business questions are answered using available data.
- Important sales, customer, product, seller, and operational patterns are identified.
- KPIs are clearly defined and consistently calculated.
- The dashboard provides useful management-level visibility.
- Findings are supported by data rather than assumptions.
- Recommendations are directly connected to identified business problems.
- The analysis helps management identify areas requiring attention.
- The final documentation is clear enough for another analyst or manager to understand the work.
- Data limitations and assumptions are transparently documented.

9. ASSUMPTIONS & CONSTRAINTS

Time Constraint

Management expects the initial analysis and reporting to be completed within a limited project timeline.

The analyst should prioritize the most important business questions rather than spending excessive time on low-value analysis.

Data Constraint

The analysis will be limited to the data provided by ShopKart.

Some business information may not be available in the datasets.

The analyst should not assume that missing information exists elsewhere unless management provides it.

Data Access

The analyst receives analytical copies of business datasets.

Direct access to production systems is not included in the project.

Business Constraint

Recommendations should be realistic for a large e-commerce marketplace and should consider operational feasibility.

Management does not expect recommendations that require unlimited resources or unrealistic changes.

Reporting Constraint

The final dashboard should be understandable to non-technical management.

Technical SQL or Python implementation details should not dominate the management presentation.

Confidentiality

The provided business data should be treated as confidential company information and should not be shared outside the project.

Assumption

The datasets represent historical business activity provided by ShopKart for analytical purposes.

The analyst is expected to validate the data before using it for business conclusions.

Important Client Expectation

We don't want a report that simply tells us what the data contains.

We want to understand what is happening in our business, where the important problems are, and what management should pay attention to.

At this stage, however, we are not asking you to provide the answers yet.

We want you, as the Data Analyst, to investigate the data and come back with evidence.